

# Age norms and compulsory schooling: United States (and Canada), 1850–1920



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At what age should a PhD student complete their dissertation?

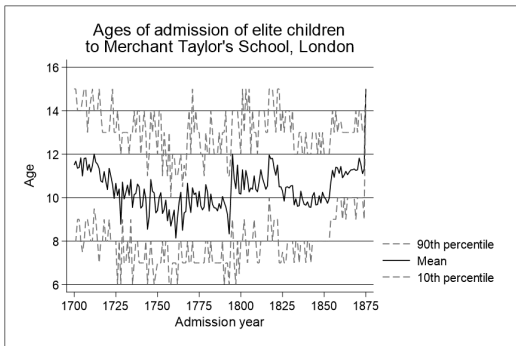
When should children learn to talk?

When is it too late to change careers?

At what age should I plan to buy a house and start a family?

- Age norms relate biological age to specific life events
- Deviating from a 'norm' is associated with social sanctions  
(Settersten 2003)
- Often directly relevant to economic outcomes
- Differences across cultures exist (Mortimer et al. 2005; Hajnal 1953)
- **How do these norms develop?** Case study in origins of norms  
(Akerlof 2020; Xue 2025)

# School-entry age



- Age a poor predictor of academic performance in the past (Ariès 1964; Vinovskis et al. 1995; Caruso 2023; Lachanski 2025)
- Learning was organised by ability, not age

- School manuals in Europe rarely recommended grouping children by age (Caruso 2023)

**Table 6.** Age as the *main* criterion for the classification of schoolchildren in classes and sessions, by country (1830–1900).

|                              | UK and Ireland | France | Spain | Total |
|------------------------------|----------------|--------|-------|-------|
| Age main criterion           | 1              | 19     | 1     | 21    |
| Age not main criterion       | 28             | 17     | 37    | 82    |
| Unclear/topic not considered | 6              | 6      | 10    | 22    |
| Total                        | 35             | 42     | 48    | 125   |

Source: Own elaboration.

- Not all countries the same (e.g. France)
- U.S. movement to standardize classrooms (c. 1840) and exclude very young and very old students (Vinovskis et al. 1995)

### **Recitation method (Early 19th century)**

Long-established pedagogy. Small groups of children given individualised instruction.

### **Monitorial/graded instruction (c. 1830-70)**

Division of teaching labour. Junior or assistant teachers given elementary subjects. Children classed by ability across subjects.

### **Age-graded instruction (c. 1900)**

Chronological age key feature of classroom structure. Proliferation of age-for-grade statistics after 1900.

### Introduction of compulsory schooling c. 1870-1920

- Primary education in 'common schools' widespread before compulsion
- Local control and funding, usually free but voluntary

### Many contemporary changes to childhood and youth:

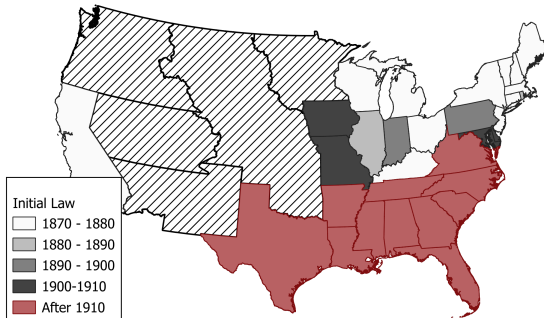
- Rise in juvenile courts (Cohen 1982)
- Reforms to age of consent
- Developmental and educational psychometrics (Vinovskis et al. 1995)
- Changes to liability and insurance (Zelizer 1994)

Compulsory schooling standardizes entry, age norm develops (Lassonde 2013) ; alternatively, a bureaucratic tool (e.g. Eisenstadt 1956)

- Age regulations existed before compulsory schooling to establish eligibility for common-school admission and distribute resources
- E.g., in 1948, New York school code gave ages 5 to 21 right to attend but compelled ages 7 to 16 (Vinovskis et al. 1995)
- School-entry minima do not appear to have been consistently enforced



Figure: Year compulsory schooling introduced by state



\*Only states incorporated before 1850 shown.

## Human capital and childhood

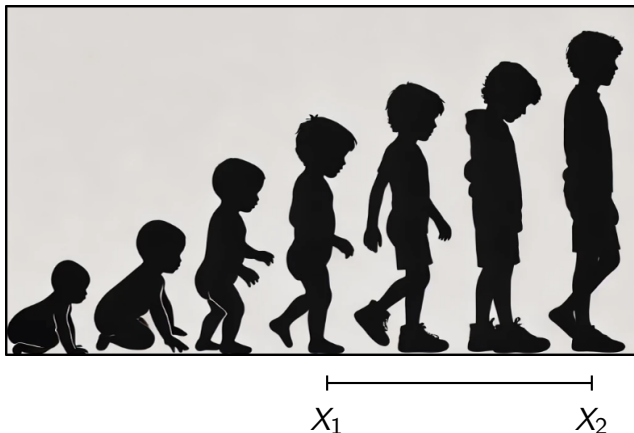
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- Education ( $X$ ) often measured as duration (e.g. Clay et al. 2021)

## Human capital and childhood

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- Can also consider entry ( $X_1$ ) and exit ( $X_2$ ) separately

## School entry

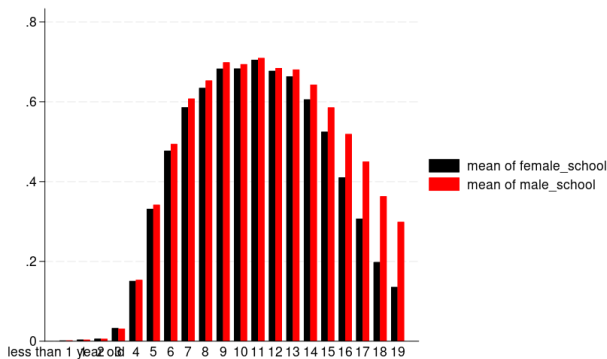
- Cognitive skills (Cunha and Heckman 2008)
- Child care (Barua 2014) vs. transaction costs
- **Culture?** (Borck 2014; El-Attar 2013)

## School exit

- Labour market (e.g. Clay et al. 2021)
- Specific skills (Goldin & Katz 2009)
- Signalling (Huntington-Klein 2021)

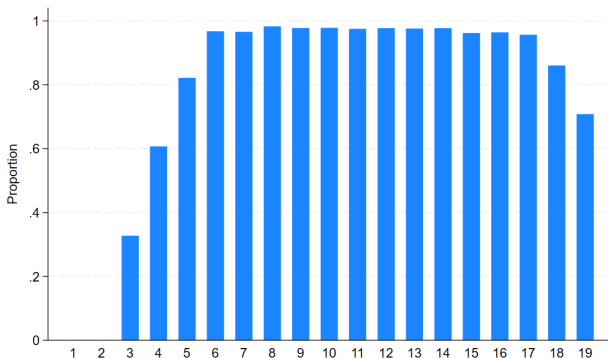
How initial compulsory schooling laws affect school-entry age largely unexplored (ex. MacKinnon & Minns 2009) .

Figure: Proportion in school by age and sex, Ohio, 1850



- In 1850, 99% enter school between 2 and 10

Figure: Proportion in school by age, Ohio, 2019



- In 2019, 99% enter school between 3 and 6

### Q1:

- What characteristics are associated with variation in school-entry age?
- In particular, do migrants exhibit meaningful differences in school-entry age?

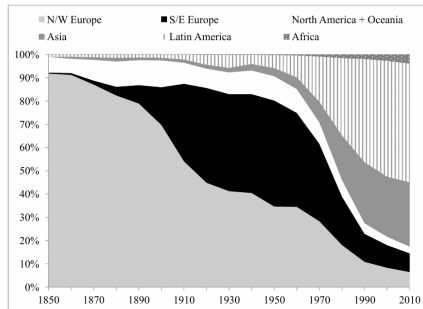
### Q2:

- Do compulsory schooling laws affect levels and variation in school-entry age across states?

- Unrestricted until 1917 (some exceptions)
- Two European 'waves'
- Migration from European periphery negatively selected

(Abramitzky & Boustan 2017)

Figure 2: Sending regions within the foreign-born population, 1850-2010

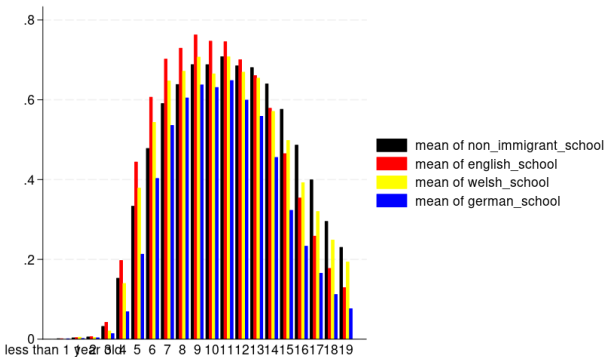


Source: Authors' calculations based on IPUMS samples of US Census (Ruggles, et al., 2010).

(Abramitzky & Boustan 2017)

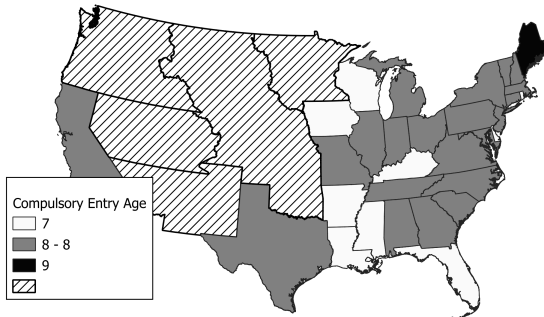


Figure: Proportion in school by age and migrant background, Ohio, 1850



- Ohio's first compulsory schooling law in 1877
- School-entry age 8 — not binding for many children

Figure: Compulsory school-entry age adopted by state



\*Only states incorporated before 1850 shown.

## ID1:

Observe determinants of school-entry age over time (descriptive)

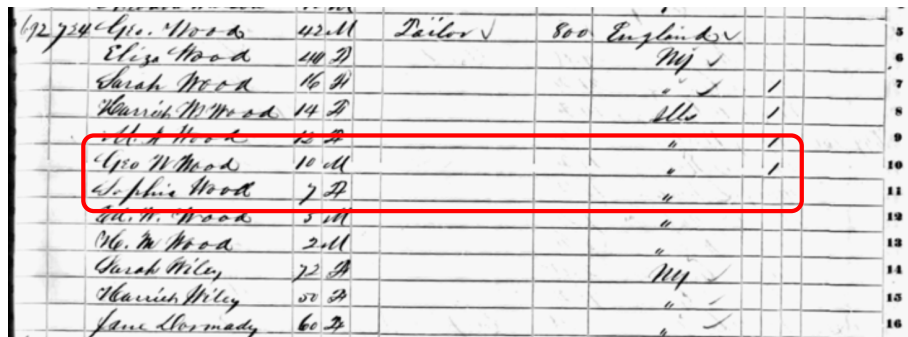
## ID2:

Staggered DiD in 1850-1920 U.S. Census data

- Rely on partial identification due to selection (Rathnayake et al. 2024)
- H1: - enforcement of law
- H2: + norm change

⚠ Entry age can be constructed from age-specific enrolment rates s.t. assumptions about school duration

## Method



A handwritten table, likely a census record, with columns for names, ages, occupations, and birthplaces. The table is numbered 1 through 16 on the right side. A red rectangular box highlights the row for 'Geo W Wood', which is row 10. The row contains the following data: 'Geo W Wood', '10 M', and '1'.

| Row | Name          | Age  | Occupation | Birthplace    | Other |
|-----|---------------|------|------------|---------------|-------|
| 5   | Geo. Wood     | 42 M | Tailor ✓   | 800 England ✓ |       |
| 6   | Eliza Wood    | 40 F |            | Nij ✓         |       |
| 7   | Sarah Wood    | 16 F |            | " ✓           | 1     |
| 8   | Harriet Wood  | 14 F |            | Wls           | 1     |
| 9   | Geo W Wood    | 12 M |            | "             | 1     |
| 10  | Geo W Wood    | 10 M |            | "             | 1     |
| 11  | Sophie Wood   | 7 F  |            | "             |       |
| 12  | Ed. W. Wood   | 5 M  |            | "             |       |
| 13  | Wm. W Wood    | 2 M  |            | "             |       |
| 14  | Sarah Wiley   | 72 F |            | Nij ✓         |       |
| 15  | Harriet Wiley | 50 F |            | " ✓           |       |
| 16  | Jane Normady  | 60 F |            | " ✓           |       |

### Key assumption:

Household school-entry lies between oldest non-student and youngest student

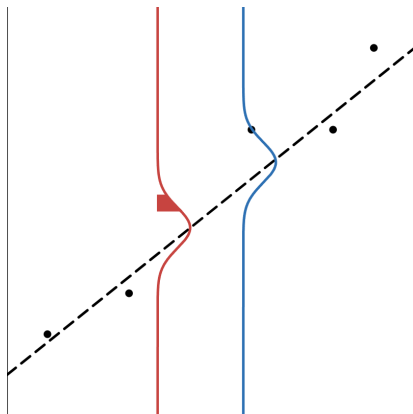
- ML selects model parameters to maximise observed data points:

$$\max \mathcal{L} = \sum_{i=0}^N f(y_i|x)$$

- This can also be done for observed intervals:

$$\max \mathcal{L} = \sum_{i=0}^N F(R_i|x) - F(L_i|x)$$

- Proceed with OLS

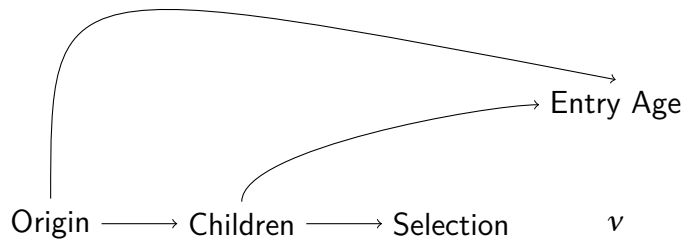


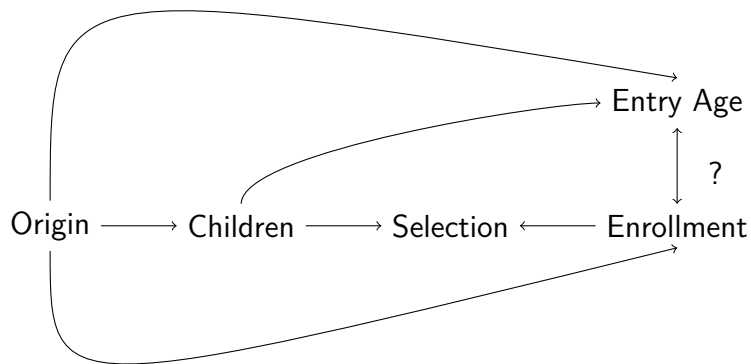
## Method

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- Look only at first family in household (91.22 %) to eliminate complex group residence
- Sort households by age, count when older child has different student status

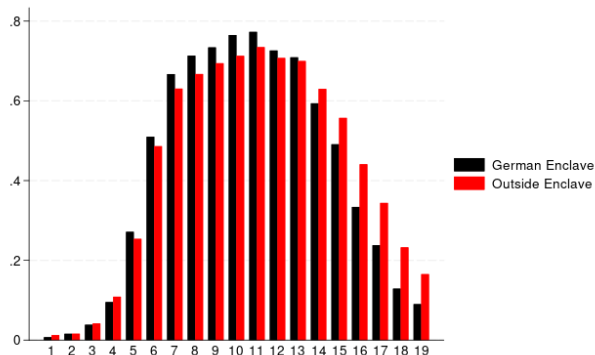
| Changes | Frequency | Percent | Cumulative |
|---------|-----------|---------|------------|
| 0       | 1,451,721 | 49.14   | 49.14      |
| 1       | 550,294   | 18.63   | 67.76      |
| 2       | 895,442   | 30.31   | 98.07      |
| 3+      | 57,020    | 1.93    | 100        |







## School enrollment of Swiss migrants in German Enclaves



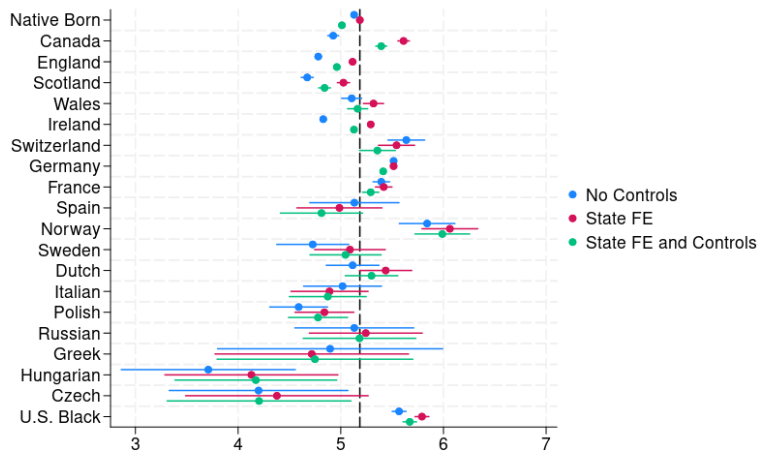
- Local control affects language of instruction
- Cannot reject null of no relationship between enrollment and entry-age ( $p = 0.55$ )

$$y_i = \alpha + \mathbf{X}\beta + \mathbf{C}\gamma + \epsilon_i$$

In a repeated cross-section of US census data (IPUMS 2025):

- $y_{is}$  is school-entry age of family  $i$  estimated using interval method
- $\mathbf{X}$  are dummy variables indicating migrant origin
- $\mathbf{C}$  are control variables including number of children and household head's occupation

## Descriptive Results: 1850



$$\ln(\sigma) = 1.23 - 0.42d_{1880} - 0.26d_{1900}$$

## International School-Entry Laws

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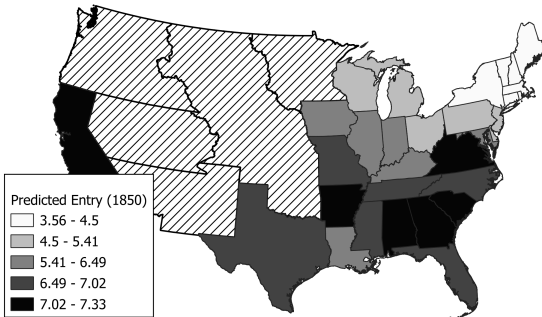
| State             | Date    | Age    | Details                                         |
|-------------------|---------|--------|-------------------------------------------------|
| United Kingdom    | 1876    | 5      | -                                               |
| Austria           | 1774    | 6      | Compulsory education, not necessarily at school |
| Prussia           | 1794    | 6      | -                                               |
| Swiss Confederacy | 1803-32 | 7      | Vaud and Solothurn Cantons                      |
| Netherlands       | 1900    | 7      | -                                               |
| Denmark           | 1814    | 6 or 7 | -                                               |
| France            | 1882    | 6      | -                                               |
| Sweden            | 1842    | 9      | School-entry age poorly defined in the act      |
| Sweden            | 1882    | 7      | School-entry age more clearly defined           |
| Italy             | 1859    | 6      | -                                               |

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Source: Westberg et al. 2019; *Allgemeines Landrecht für die Preußischen Staaten*, 1794.

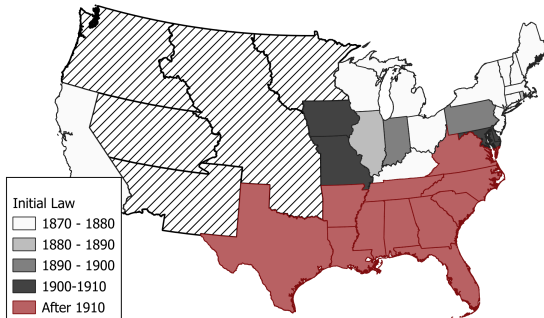
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Figure: Expected school-entry age of non-migrants in 1850 by state



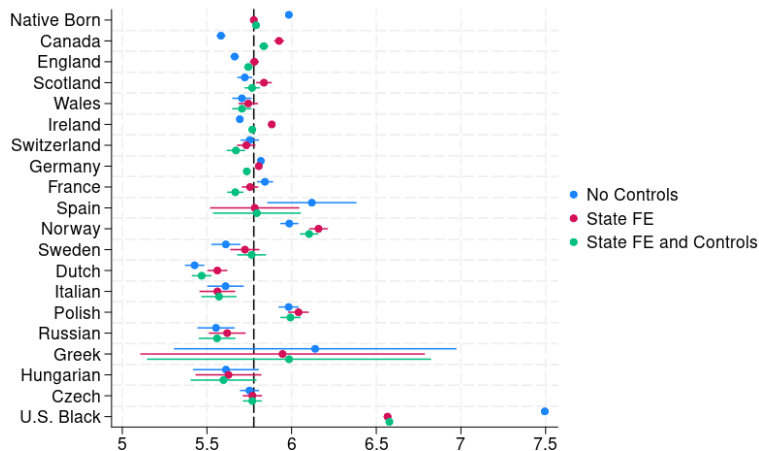
\*Only states incorporated before 1850 shown.

Figure: Year compulsory schooling introduced by state



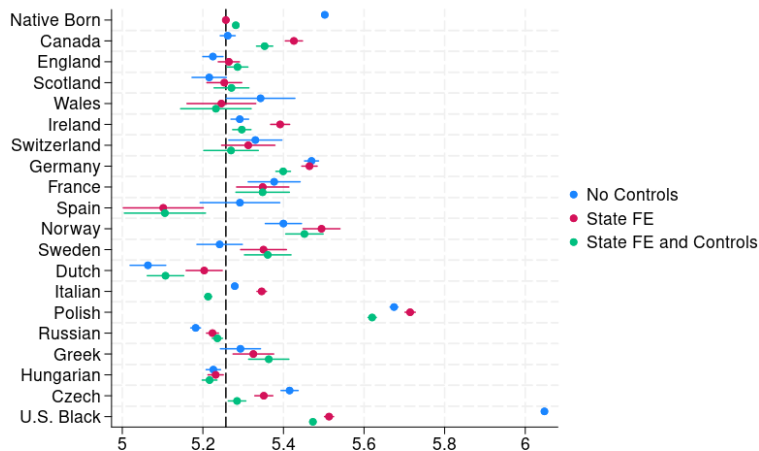
\*Only states incorporated before 1850 shown.

## Descriptive Results: 1880



$$\ln(\sigma) = 1.03 - 0.16d_{1880} - 0.17d_{1900}$$

## Descriptive Results: 1920



$$\ln(\sigma) = 0.71 - 0.02d_{1880} - 0.07d_{1900}$$



## Model 2: Staggered DiD

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Wooldridge (2022) characterises TWFE as specification error when treatment is staggered, proposes

$$y_{it} = \eta + \lambda_q d_{iq} + \dots + \lambda_T d_{iT} + \sum_{s=2}^T \theta_s f_{st} + \sum_{r=q}^T \sum_{s=r}^T \tau_{rs} (d_{ir} \cdot f_{st})$$

where

- $q \dots T$  are treatment periods
- $d_i$  are treatment 'cohort' indicators
- $f_{st}$  are time period indicators
- And treatment effects are estimated for each post-treatment period

## Model 2: Staggered DiD

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$$y_{it} = \eta + \lambda_q d_{iq} + \dots + \lambda_T d_{iT} + \sum_{s=2}^T \theta_s f_{st} + \sum_{r=q}^T \sum_{s=r}^T \tau_{rs} (d_{ir} \cdot f_{st}) + \\ \mathbf{x}_i \kappa + \sum_{r=q}^T (d_{ir} \cdot \mathbf{x}_i) \zeta_r + \sum_{s=2}^T (f_{st} \cdot \mathbf{x}_i) \pi_s + \sum_{r=q}^T \sum_{s=r}^T (d_{ir} \cdot f_{st} \cdot \overline{\mathbf{x}_r}) \rho_{rs}$$

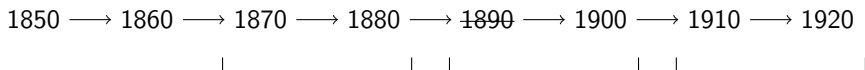
where controls  $\mathbf{x}$  are included with a full set of interactions and are demeaned,  $\overline{\mathbf{x}_r}$ , when interacted with the TEs to ease interpretation.

Literature suggests several relevant controls:

- Women's wages (Henderson 2024)
- Number of migrants in state (Lleras-Muney & Shertzer 2015; Bandiera et al. 2018)
- Size of Black population (Engerman & Sokoloff 2012)
- Inequality (Goñi 2023; Galor & Moav 2006)
- Union admission date (e.g. Paglayan 2024)

## Model 2: Data limitations

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Adopt two treatment cohorts:  $D_0$  for 1870-1880 and  $D_1$  for 1890-1900:

- Clustered standard errors at state-year level (e.g. Clay et al. 2021) cannot otherwise be calculated
- Few states adopt compulsion between 1880-1900; few degrees of freedom for state-level controls
- 'Not-yet-treated' as control group
- Southern states fully interacted with treated years

## Model 2: Controls

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Keep the following state-level controls:

- Women's wages (Henderson 2024)
- Number of migrants in state (Lleras-Muney & Shertzer 2015; Bandiera et al. 2018)
- Legislated entry-age

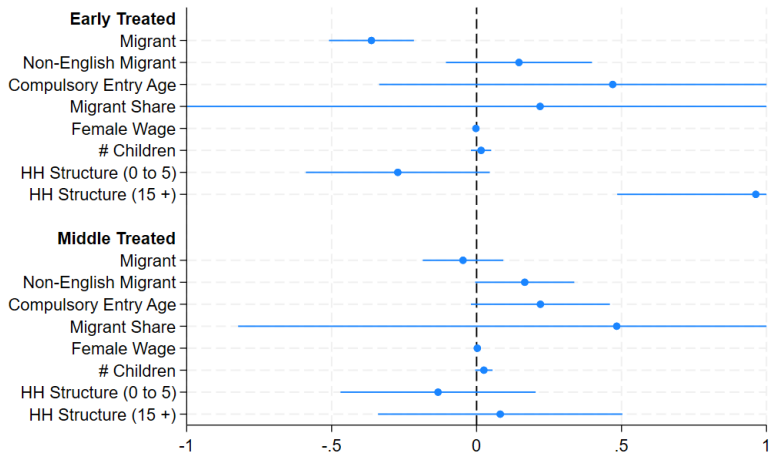
Additional household-level controls:

- No. of children
- Proportion of children aged 0-5
- Proportion of children aged 15 +
- Occupational score of household head
- Migrant
- Non-English-speaking migrant

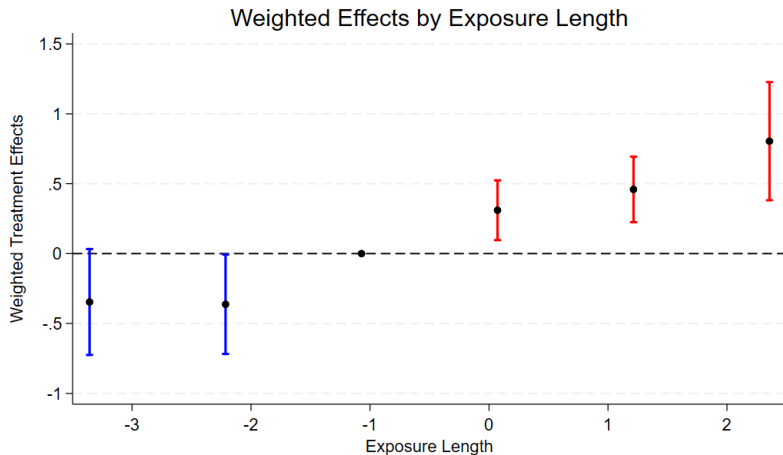
|                             | (1)<br>OLS: Entry Age | (2)<br>OLS: Entry Age |
|-----------------------------|-----------------------|-----------------------|
| <b>Treatment Effects</b>    |                       |                       |
| $D_0 \times F_{1880}$       | 0.50 (0.22)           | 0.39 (0.16)           |
| $D_0 \times F_{1900}$       | 0.79 (0.34)           | 0.72 (0.18)           |
| $D_0 \times F_{1920}$       | 0.97 (0.25)           | 0.77 (0.18)           |
| $D_1 \times F_{1900}$       | 0.26 (0.34)           | 0.19 (0.11)           |
| $D_1 \times F_{1920}$       | 0.38 (0.24)           | 0.28 (0.10)           |
| <b>Treatment Cohort FEs</b> |                       |                       |
| $D_0$                       | -1.10 (0.22)          | 4.90 (3.27)           |
| $D_1$                       | -0.19 (0.19)          | 8.47 (.)              |
| <b>Time Dummies</b>         |                       |                       |
| $F_{1880}$                  | 0.32 (0.16)           | -0.66 (0.87)          |
| $F_{1900}$                  | 0.32 (0.30)           | -0.79 (.)             |
| $F_{1920}$                  | -0.65 (0.22)          | -2.48 (.)             |
| <b>South Controls</b>       |                       |                       |
| south                       | 1.19 (0.19)           | 1.21 (0.15)           |
| south $\times F_{1880}$     | -0.042 (0.19)         | 0.17 (0.14)           |
| south $\times F_{1900}$     | 0.11 (0.33)           | 0.13 (0.18)           |
| south $\times F_{1920}$     | -0.46 (0.24)          | -0.33 (0.16)          |
| Constant                    | 5.82 (0.18)           | 3.29 (.)              |
| Controls                    | NO                    | YES                   |
| Log-Likelihood              | -1397738.2            | -1259582.1            |
| Observations                | 1962489               | 1846100               |

Clustered standard errors in parentheses.

## DiD: TE interactions



## DiD: Pre-trends and weighted effect





## Partial Identification: Selection Bias

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|                  | Treatment Effects     |                       |                       |                       |                       |
|------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
|                  | $D_0 \times F_{1880}$ | $D_0 \times F_{1900}$ | $D_0 \times F_{1920}$ | $D_1 \times F_{1900}$ | $D_1 \times F_{1920}$ |
| Selection Effect | -0.03                 | -0.04                 | -0.02                 | -0.02                 | 0.0                   |

- Rathnayake et al. 2024 provide method to bound TE in case of selection for always-observed group
- In presence of sample selection, DiD estimator is biased:

$$\tau = q_{OO11} \cdot \tau_{OO} + (1 - q_{OO11}) \cdot \tau_{NO} + (1 - q_{OO11}) \cdot (E[Y_1(0)|D=1, t=1, NO] - E[Y_1(0)|D=1, T=1, OO])$$

- Where  $q_{0011}$  is the share of always-observed group in post-treatment period and final term is difference in counterfactual

## DiD: Migrant school-entry age

|                             | (1)<br>Probit: Early Entry | (2)<br>Probit: Early Entry |
|-----------------------------|----------------------------|----------------------------|
| <b>Treatment Effects</b>    |                            |                            |
| $D_0 \times F_{1880}$       | -0.24 (0.09)               | 0.031 (0.10)               |
| $D_0 \times F_{1900}$       | -0.31 (0.13)               | -0.13 (0.13)               |
| $D_0 \times F_{1920}$       | -0.21 (0.11)               | -0.075 (0.14)              |
| $D_1 \times F_{1900}$       | -0.11 (0.12)               | -0.040 (0.07)              |
| $D_1 \times F_{1920}$       | -0.055 (0.09)              | -0.013 (0.08)              |
| <b>Treatment Cohort FEs</b> |                            |                            |
| $D_0$                       | 0.37 (0.10)                | -2.85 (1.71)               |
| $D_1$                       | 0.054 (0.09)               | -4.31 (0.67)               |
| <b>Time Dummies</b>         |                            |                            |
| $F_{1880}$                  | -0.34 (0.06)               | 0.53 (0.67)                |
| $F_{1900}$                  | -0.20 (0.11)               | 0.052 (0.58)               |
| $F_{1920}$                  | -0.58 (0.09)               | 0.098 (0.64)               |
| <b>South Controls</b>       |                            |                            |
| south                       | -0.43 (0.08)               | -0.36 (0.10)               |
| south $\times F_{1880}$     | 0.28 (0.07)                | 0.087 (0.08)               |
| south $\times F_{1900}$     | 0.29 (0.11)                | 0.16 (0.11)                |
| south $\times F_{1920}$     | 0.42 (0.10)                | 0.25 (0.12)                |
| Constant                    | -2.07 (0.08)               | -1.64 (0.51)               |
| Controls                    | NO                         | YES                        |
| Log-Likelihood              | -228194.1                  | -194691.4                  |
| Observations                | 3816115                    | 3549063                    |

Clustered standard errors in parentheses.

- Compulsory schooling in U.S. resulted in *later* school-entry age
- This complicates impact of the law due to potential impact on *human capital, labour supply*, etc. (e.g. Clay et al. 2021)
- Law likely non-binding suggests shifting norm (Lassonde 2013)
- Convergence in age norm across cultures

What next?

- Do British migrants to Canada also converge to local school-entry age?

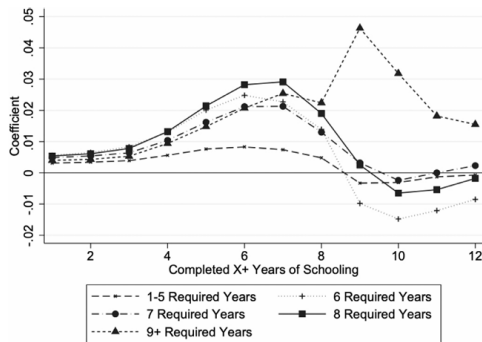
Address known shortcomings:

- Include 1870 for more pre-treatment observations
- Address measurement error using linked samples
- Look more closely at enforcement of school-entry minima

# Thank you!

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<http://louisstevenhenderson.github.io>



**Fig. 2.** Effect of the Laws on Completed Years of Schooling, Native-Born White Men

Notes: Figure shows the coefficients from regressing completed years of schooling on laws using individuals who were born between 1885 and 1912, inclusive. The confidence intervals on each point can be obtained from Appendix [Table 1](#), which shows the complete regression estimates. The outcome in each regression is a binary indicator for having completed at least as many years as shown on the axes. Results are shown for the all white men sample that has 19,055,381 observations. Each regression includes state of birth and region by year of birth fixed effects. See Appendix [Table 1](#) for full results.